

NAMANDEEP SINGH NAYYAR

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SUMMARY

Applied AI and Data Analytics professional experienced in building automation-driven data platforms, operational intelligence systems, and real-time decision support tools. Hands-on experience working with large datasets, developing machine learning workflows, and designing analytics solutions that improve planning efficiency, process visibility, and business performance. Interested in solving product, growth, and operational challenges through a scalable data-driven system.

EDUCATION

Babson College | *Master of Science, Business Analytics*

Boston, USA Aug 2024 - Dec 2025

DY Patil University | *Bachelor of Technology, Bioinformatics and Data Science*

Mumbai, India Jun 2019 - Jul 2023

TECHNICAL SKILLS

- **Languages:** Python, R, SQL, JavaScript
- **Machine Learning & AI:** NumPy, Pandas, PyTorch, scikit-learn, XGBoost, Hugging Face, LLMs, Computer Vision, Feature Engineering, Model Evaluation Metrics, Performance Monitoring, Retrieval Augmented Generation (RAG), Prompt Engineering, AI Workflow Design, LangChain, Transformers
- **Data Engineering & Systems:** Data Pipelines, REST APIs, FastAPI, PostgreSQL, Workflow Automation, System Integration, ETL / ELT Concepts, Vector Databases, Data Modeling
- **Analytics & Business Intelligence:** A/B Testing, KPI Design, Business Metrics Analysis, Forecasting
- **Visualization Tools:** Tableau, Power BI, seaborn, Streamlit
- **Developer Tools:** Git, Google Cloud Platform, Model Deployment, API Deployment, Cloud Deployment, AWS, Excel (Advanced), Docker, SEO

EXPERIENCE

ReDiPath

May 2025 - Aug 2025

Artificial Intelligence and Data Science Intern - Digital Pathology

Boston

ReDiPath | Python, PyTorch, Deep Learning, CNNs, Transformers, Image Analysis

- Developed reusable preprocessing and validation pipelines for large pathology imaging datasets to support CNN-based experimentation workflows.
- Performed exploratory analysis and structured data quality checks to improve dataset reliability for model training and evaluation.
- Prepared training and testing datasets and tracked validation performance metrics across multiple experimentation cycles.
- Collaborated with engineering and clinical teams to document dataset transformations, model assumptions, and performance observations.

Unity Enterprise

Aug 2022 - Aug 2024

Business Intelligence Analyst

India

Unity Enterprise | Python, SQL, Tableau, Excel, ARIMA, Regression, SEO, Forecasting, Data Automation

- Managed end-to-end analytics and reporting workflows supporting operations across multi pharmacy retail locations.
- Automated SQL-driven reporting pipelines connected to live dashboards, reducing manual reporting effort and improving data reliability.
- Maintained clean, well-structured datasets by standardizing data definitions, resolving inconsistencies, and performing routine QA checks.
- Built forecasting and trend analysis models to support inventory planning and operational decision-making.
- Partnered with operations and marketing teams to translate data insights into actionable process improvements.

Indian Institute of Management Indore, India

Sep 2023 - Mar 2024

Management Trainee – Executive Program in General Management

Indore

- Applied finance, marketing, and analytics concepts to develop a data-driven market entry and growth strategy for India's lab-grown diamond industry, presented to faculty and industry experts.

PROJECTS

Conversational Lead Qualification & Sales Automation System | <https://tinyurl.com/2vpe5nr9>

- skills: Machine Learning, AI Automation / LLM Applications, Data Analytics, Data Visualization, API Integration & Workflow Automation
- Built a conversational lead qualification system that captures natural language interactions and converts them into structured sales pipeline data.
- Developed configurable workflow logic for pricing estimation, lead scoring, and automated appointment scheduling across web channels.
- Designed centralized lifecycle tracking for conversations, quotes, and bookings with extensibility for CRM and multi channel deployment

Real Time Warehouse Operations & Inventory Intelligence System | <https://tinyurl.com/y823aj62>

Dec 2025 - Jan 2026

- Skills: Next.js, React, Python, FastAPI, REST APIs, PostgreSQL (Supabase), QR Codes, CCTV Systems, Event-Based Modeling, Inventory State Management, Git, Deployment (Railway)
- Designed and deployed an AI-powered warehouse inventory intelligence system that tracks real-time box movement using camera-based QR scanning and event-driven state management.
- Developed smart storage recommendation logic that optimizes location assignment based on operational parameters.
- Designed live analytics dashboards and data pipelines that synchronize physical warehouse activity with digital inventory records, improving visibility, planning efficiency, and operational decision making

AI Driven Invoice Intelligence & Automation Platform | <https://tinyurl.com/bdh7xkyl>

- Skills: Python, OCR, Machine Learning, Data Pipelines, FastAPI, PostgreSQL, Dashboarding, Workflow Automation
- Designed an end to end invoice intelligence system that converts unstructured financial documents into validated, analytics ready datasets for operational decision support.
- Built OCR driven data pipelines for automated extraction, cleansing, anomaly detection, and workflow orchestration across finance processes.
- Developed scalable backend services and interactive dashboards to track vendor performance, payment timelines, and spending patterns.
- Reduced manual processing dependency by enabling structured financial visibility and real time reporting capabilities

Personal Data Automation & Financial Intelligence Platform | <https://tinyurl.com/9rpmaar2>

- Skills: PWA Development, React, FastAPI, Data Pipelines, Email Parsing, PostgreSQL, Dashboarding, Workflow Automation
- Designed and developed a Progressive Web App (PWA) that enables users to capture personal and financial activity data through manual inputs and automated email based extraction workflows.
- Built data ingestion pipelines to parse incoming emails, structure information, and update centralized analytics dashboards in real time.
- Implemented responsive mobile first interfaces and backend APIs to support cross device usage, workflow automation, and continuous reporting visibility.
- Reduced manual tracking effort and improved decision readiness by enabling automated data consolidation and performance monitoring